Scalar fields and split torus families

Human Compression project

Abstract

One elementary divisibility identity determines all intersections in a family of Suzuki coset pregeometries. The same identity proves that the cyclic scalar group in each affine residue acts irreducibly. This gives a uniform proof of the intended families in Constructions 5 and 7 of the fixed higher-rank source, after stating its necessary indexing and order corrections explicitly. Only residual weak primitivity is asserted here.

1 The arithmetic of the scalar subgroup

Let $e \geq 3$ be odd, let $k \geq 1$, let $a_1, \ldots, a_k$ be distinct odd primes, and put

$$t = 2^e, \quad A = \prod_{i=1}^{k} a_i, \quad q = t^A.$$

Choose distinct primes ℓ_i such that

$$\ell_i \mid \frac{t^{a_i} - 1}{t - 1}, \qquad M = (t-1) \prod_{i=1}^{k} \ell_i.$$

The primes ℓ_i need not be coprime to $t - 1$. For $J \subseteq \{1, \ldots, k\}$ set

$$D_J = \prod_{i \notin J} a_i, \quad q_J = t^{D_J}, \quad M_J = (t-1) \prod_{i \notin J} \ell_i.$$

Lemma 1. *For every J one has $\gcd(M, q_J - 1) = M_J$. In particular $M \mid q - 1$. The multiplicative subgroup of order M_J in $\mathbb{F}_{q_J}^*$ generates $\mathbb{F}_{q_J}$ as a field over $\mathbb{F}_2$.*

Proof. If $\ell_i \nmid t - 1$, the order of t modulo ℓ_i is a_i: it divides that prime and is not one. Hence $\ell_i \mid t^{D_J} - 1$ exactly when $i \notin J$. Its exponent in M is one, since the chosen ℓ_i are distinct.

If $\ell_i \mid t - 1$, reduction of $1 + t + \cdots + t^{a_i - 1}$ modulo ℓ_i shows that $\ell_i = a_i$. For an odd prime $\ell \mid t - 1$, successive binomial expansions give

$$v_\ell(t^D - 1) = v_\ell(t - 1) + v_\ell(D).$$

For completeness, raising a number congruent to one modulo ℓ to a prime power step r preserves the valuation when $r \neq \ell$: the first binomial term has coefficient r, and all later terms have larger valuation. When $r = \ell$, the first term gains exactly one power of ℓ and all later terms have strictly larger valuation, since ℓ is odd. Iterating proves the displayed formula. Since D_J is squarefree, the additional factor ℓ_i in M occurs in $q_J - 1$ precisely when $a_i \mid D_J$. All other prime powers of $t - 1$ occur on both sides. This proves the gcd identity.

Suppose that the subgroup of order M_J is contained in a subfield $\mathbb{F}_{2^f} \subseteq \mathbb{F}_{q_J}$. Its subgroup of order $2^e - 1$ gives $2^e - 1 \mid 2^f - 1$, so $e \mid f$ by the usual gcd identity for powers of two. Write $f = eD$, where $D \mid D_J$. Applying the identity just proved to $t^D - 1$ shows that each a_i with $i \notin J$ divides D; otherwise its additional prime factor in M_J would be missing. Thus $D = D_J$, and the subfield is the whole field. □

2 All subgroup intersections and the maximality condition

We use the classical Suzuki matrix model, its standard subfield groups, and maximality of prime-relative-degree inclusions of simple Suzuki groups as attributed inputs. Write $S_d = \mathrm{Sz}(2^d)$ for a standard subfield group, Z_{2^d} for the involution centre of its standard Sylow two-subgroup, and H_r for the subgroup of order r in the ambient split torus. The standard Weyl involution w inverts that torus. Unique matrix coordinates give

$$S_b \cap S_c = S_{\gcd(b,c)}.$$

Conjugation by the split torus on Z_q is multiplication by a power coprime to $q-1$. These coordinate and action facts have complete derivations, with the classical ambient model explicit, in the acquired Suzuki preliminary and normalizer notes. We do not assume the source's classification of all residually weakly primitive pregeometries.

In $G = S_{eA}$ consider the following $k+2$ parabolics, with separate labels for the last two:

$$P_i = S_{eA/a_i}\ (1 \le i \le k), \qquad P_D = H_M \rtimes \langle w \rangle, \qquad P_E = Z_q \rtimes H_M.$$

For a set of selected subfield labels J, their intersection is S_{eD_J}. Intersecting further with either or both of the last two parabolics gives exactly

$$\begin{aligned} S_{eD_J} \cap P_D &= H_{M_J} \rtimes \langle w \rangle, \\ S_{eD_J} \cap P_E &= Z_{q_J} \rtimes H_{M_J}, \\ S_{eD_J} \cap P_D \cap P_E &= H_{M_J}. \end{aligned}$$

Indeed the diagonal parameter must belong to $\mathbb{F}_{q_J}$, so Lemma 1 gives its precise cyclic order. The central unipotent parameter is restricted to that same field. The reflection coset of the split normalizer is outside the point stabilizer, leaving only the torus in the third intersection. This also proves the formulas when J is empty.

Theorem 2. *The displayed parabolics define a residually weakly primitive coset pregeometry of rank $k+2$. Its full intersection is H_{t-1}.*

Proof. The condition is that every proper subset of the labels admits a remaining parabolic whose intersection is maximal proper in the current intersection. We check all four possibilities concerning the labels D, E.

If neither has been selected and a subfield label remains, adjoining any such label gives a prime-relative-degree inclusion $S_{eD_J/a_i} < S_{eD_J}$. Both are simple, since their exponents are at least $e \ge 3$, so this is maximal. If no subfield label remains, the current intersection is S_e, and adjoining D gives its full split torus normalizer, a classical maximal subgroup.

If D but not E has been selected, adjoining E passes from $H_{M_J} \rtimes \langle w \rangle$ to its cyclic subgroup of index two.

If E but not D has been selected, adjoining D passes from $Z_{q_J} \rtimes H_{M_J}$ to H_{M_J}. This is maximal because the scalar action is irreducible over $\mathbb{F}_2$. To verify this last assertion, an additive subspace invariant under the cyclic scalar group is invariant under the field it generates. Lemma 1 identifies that field with $\mathbb{F}_{q_J}$, whose one-dimensional vector space has only the zero and full subspaces. Every intermediate subgroup containing H_{M_J} is the product of that complement and its intersection with Z_{q_J}; hence no proper nontrivial intermediate subgroup exists.

Finally, if both D, E have been selected, a subfield label i remains because the chosen subset is proper. Adjoining it passes from H_{M_J} to H_{M_J/ℓ_i}, of prime index ℓ_i. This exhausts the possibilities and proves the condition. The intersection formulas give the stated full intersection. □

3 The two source families and their literal corrections

The source is D. Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20. We use its fixed twenty-page edition. The introduction to the constructions, on p. 15, instructs the reader to retain the intended whole intersection lattice from the rank-four discussion. We therefore distinguish this intended lattice from literal indexing or order errors in the displayed recipes.

For Construction 7 put $k = \alpha - 2$, and take its selected relative prime degrees as $a_1, \ldots, a_k$. Its definition of t gives our $e = \log_2 t$. Its range $\alpha \leq \sum_j e_j$ leaves at least two prime factors, with multiplicity, in e, so $e \geq 3$. Its cyclic parameters are our ℓ_i. The two displayed soluble group orders are exactly those in Theorem 2. The source labels the selected Suzuki groups $G_1, \ldots, G_{\alpha-2}$ and then also labels the dihedral group $G_{\alpha-2}$, while its final tuple begins with G_0. Relabel the subfield groups $P_1, \ldots, P_k$ and keep D, E as separate labels. With that explicit indexing correction, every permitted parameter is realized, with all intended intersections and the maximality condition proved above.

Construction 5 has the same family with e prime. In its first case, the source exponent is a product of distinct primes. The distinguished prime is $e = p_i$, and the other $m - 1$ primes are our selected a_j; thus $k = m - 1$ and the rank is $m + 1$. In its second case, one prime p_m is repeated in the ambient exponent. Take $e = p_m$ and select all m distinct primes as the a_j, including p_m itself. They remain pairwise distinct; no coprimality between e and A is needed anywhere in the proof. The rank is $m + 2$.

The literal recipes in Construction 5 require more than relabeling. Both displayed dihedral orders omit the factor $t - 1$. Its first case assigns conflicting indices. Its second case uses an undefined i in the affine product and ends its tuple at G_m despite listing G_{m+1}. The corrected dihedral and affine groups are, in both cases, D_{2M} and $Z_q \rtimes H_M$, with the product taken over all selected indices. These are the groups required by the stated rank-four intersections in the preceding discussion. We prove this corrected family, not the inconsistent literal assignment.

The method extends the previously retained rank-three split family: subfield coordinates give the intersections, and scalar irreducibility proves the affine maximality step. The uniform arithmetic lemma supplies the higher-rank argument, including the case $\ell_i \mid t - 1$. No priority claim is made for this elementary arithmetic or for the classical Suzuki subgroup structure. Arbitrary relative positions, flag transitivity, residual connectedness, a classification of all families, and compression measurements are outside this component.